Generational Depth Estimation of Recursive Synthetic Text: Measuring the Discrimination Boundary

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Abstract

Web-scale corpora are increasingly contaminated by recursive synthetic content, yet no method estimates a document's generational depth. We measure a lower bound on the generational discrimination boundary, $K^*_{\text{empirical}}$ (the maximum depth at which adjacent generations remain distinguishable). Under prompt-chain generation across 10 model configurations (6 architecture families, 1B–8B), 4 source domains, and 4 decoding strategies, $K^*_{\text{empirical}} \leq 2$ for all web text configurations across six architecture families at 1B–8B scale (θ=0.60); structured academic text (arXiv) reaches $K^*_{\text{empirical}}$=3 across three independent architecture families. A single generation pass suffices to render general web text indistinguishable from deeper synthetic content, a *one-wash laundering* effect that concentrates all exploitable signal at the human-vs.-machine boundary. Across continuation-mode generators, generator-agnostic statistical features enable 80–92% cross-model transfer with scorer-invariant feature rankings ($\rho > 0.90$), supporting depth estimation on corpora from unknown generators without retraining. Under extreme contamination (100% synthetic), $K^*_{\text{empirical}}$ correctly predicts that binary filtering outperforms graduated alternatives; at realistic ratios ($\leq$50%), all strategies converge ($p > 0.12$).

1 Introduction

Web-scale text corpora are increasingly contaminated by synthetic content that has itself been generated from earlier synthetic content. Estimates suggest that a substantial fraction of online text is machine-generated (Sun et al., 2025; Spennemann, 2025), and this share continues to rise. When models train on the outputs of earlier models, the resulting multi-generational synthetic text degrades with each iteration: tail modes vanish, lexical diversity contracts, and surprisal distributions flatten (Shumailov et al., 2024; Dohmatob et al., 2024). This degradation arises in both *retrain-chains* (iteratively training on model outputs) and *prompt-chains* (recursively generating from previous outputs without retraining); we study the latter, which isolates generation-level information loss from training dynamics (Appendix B.7). Yet the dominant paradigm treats all machine-generated text identically, discarding the generational depth dimension.

This gap cuts across four lines of work: binary detectors (Basani and Chen, 2026; Zhang et al., 2025) produce no depth estimate (Section 4.6); membership inference (Zawalski et al., 2026) addresses a different question; the model collapse literature (Shumailov et al., 2024; Alemohammad et al., 2024; Schaeffer et al., 2025) provides no per-text tool; and data selection methods (Xie et al., 2023b; Tirumala et al., 2023; Drayson et al., 2025) ignore recursive depth. No existing method estimates how many rounds of recursive generation a text has undergone.

The Data Processing Inequality guarantees that information degrades monotonically across generations, motivating a finite discrimination boundary K^*. We empirically measure a lower bound $K^*_{\text{empirical}}$ on the discrimination boundary (the maximum depth at which a classifier reliably distinguishes adjacent generations) using 15 statistical features and an Ordinal Binary Decomposition (OBD) LightGBM classifier. We operationalize $K^*_{\text{empirical}}$ as the deepest $k \geq 1$ where the pairwise $\text{AUC}_{k-1,k}$ exceeds $\theta = 0.60$ (Section 3.1).

Our contributions:

(a) **Measuring the discrimination boundary**: $K^*_{\text{empirical}} \leq 2$ across all 10 model configurations (6 architecture families at 1B–8B scale) and 4 decoding strategies; $K^*_{\text{empirical}}$=2 for 6/10 spanning three families ($p \leq 0.002$; Section 4.1). A single generation pass renders web text statistically indistinguishable from deeper synthetic content; this finding es-

tablishes binary sufficiency by measurement, reveals domain-dependent variation (arXiv $K^*_{\text{empirical}}$=3), and quantifies boundary sensitivity to threshold, scorer, and scale. Both supervised OBD and zero-shot LLMs fail beyond $K^*_{\text{empirical}}$ (McNemar $p > 0.05$; Section 6).

(b) **Domain dependence of $K^*_{\text{empirical}}$**: general web text (C4, Wikipedia) and code yield $K^*_{\text{empirical}}\leq 2$, while structured academic text (arXiv) yields $K^*_{\text{empirical}}=3$, consistent across model families (Section 4.5).

(c) **Generator-agnostic cross-model transfer** (within continuation mode): 15 statistical features enable 80–92% accuracy retention with scorer-invariant feature rankings ($\rho > 0.90$), supporting depth estimation on corpora from unknown continuation-mode generators without retraining; $K^*_{\text{empirical}}$ agreement holds across scorers for strong-signal configurations (Section 4.2).

As a consistency check under extreme (100%) contamination, $K^*_{\text{empirical}}$=2 correctly predicts the strategy ordering: binary curation minimizes MMLU degradation (0.5941 vs. base 0.6018); at realistic ratios ($\leq$50%), all strategies converge (Section 5).

2 Related Work

Model collapse and recursive degradation. Iterative training on model outputs causes tail-distribution degradation (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024, 2025; Fu et al., 2025), diversity contraction (Guo et al., 2024; Padmakumar and He, 2024; Briesch et al., 2023), and predictable convergence (Schaeffer et al., 2025; Kazdan et al., 2025; Gerstgrasser et al., 2024; Wang et al., 2025). Our prompt-chain operates in the replace regime (Kazdan et al., 2025), making $K^*_{\text{empirical}} = 2$ a conservative lower bound. Markov chain convergence theory predicts that recursive generation converges to a stationary distribution (Fu et al., 2025), complementing the DPI's mutual information monotonicity; empirically, detection accuracy erodes with each generation (Hao et al., 2025), consistent with our measured discrimination boundary. This literature characterizes macro dynamics but provides no per-text depth estimation, the gap we address.

Beyond binary detection. Binary detectors (Mitchell et al., 2023; Bao et al., 2024; Hans et al., 2024; Basani and Chen, 2026; Zhang et al., 2025) achieve strong human-vs.-machine separation but produce no depth estimate; other systems such as Ghostbuster (Verma et al., 2024) and GPTZero (Liang et al., 2023), as well as benchmarks like RAID (Dugan et al., 2024), likewise target the binary task without estimating generational depth. Recent work extends beyond binary labels to collaboration degree (Cheng et al., 2025), contamination hierarchies (Mehta, 2025), and multigenerator benchmarks (Wang et al., 2024); robustness concerns arise under paraphrasing (Sadasivan et al., 2023; Krishna et al., 2023) and cross-mode transfer (Hao et al., 2025); CoDeC (Zawalski et al., 2026) addresses contamination via membership inference. None estimates recursive generation depth.

Data selection and curation. DSIR (Xie et al., 2023b), D4 (Tirumala et al., 2023), and DoReMi (Xie et al., 2023a) are depth-agnostic; Drayson et al. (Drayson et al., 2025) combined binary detection with importance resampling, a special case when $K^* = 1$.

Ordinal classification and probing. Our OBD classifier follows Frank & Hall's (Frank and Hall, 2001) binary decomposition; our neural probe extends probing (Alain and Bengio, 2017; Belinkov, 2022) to the discrimination boundary. An orthogonal line of work embeds provenance signals proactively via text watermarking (Kirchenbauer et al., 2023); our work addresses the complementary post-hoc scenario where the generator is not controlled.

3 Method

3.1 Problem Formulation

A generation operator G_δ (LLM + decoding strategy δ) produces successive generations $X \to Y_1 \to \cdots \to Y_K$, where $Y_k = G_\delta(Y_{k-1})$ and $Y_0 \triangleq X$ (Figure 1). This sequence forms a Markov chain. By the Data Processing Inequality, $I(X;Y_1) \geq I(X;Y_2) \geq \cdots \geq 0$: mutual information with the seed text degrades monotonically. Together with Markov chain convergence, this motivates the existence of a finite discrimination boundary where $\text{AUC}_{k-1,k}$ falls below a decision threshold, though AUC and mutual information are not formally equivalent; the actual saturation depth must therefore be measured empirically. Pairwise AUC may exhibit non-monotonicity at deeper depths due to classifier and feature interactions (e.g., Pythia-1.4B greedy 4v5 AUC = 1.0 vs.

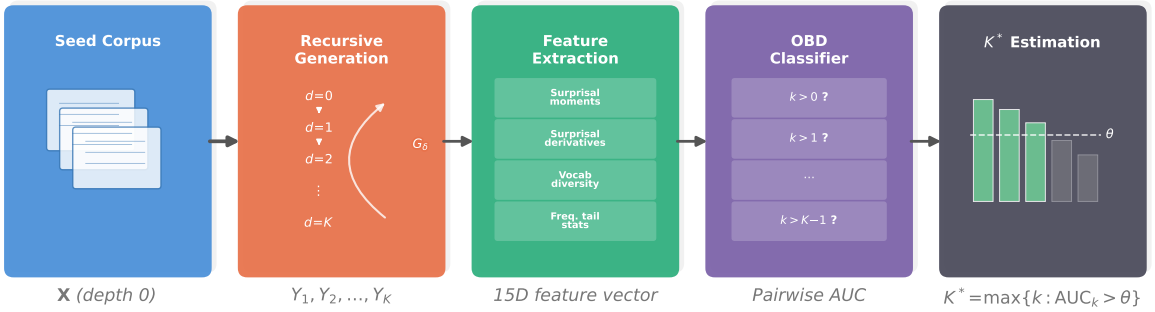


Figure 1: Generational depth estimation pipeline. Seed documents are recursively passed through a generation operator G_δ to produce a depth chain. Corpus-level statistical features are fed to the OBD-LightGBM classifier for depth prediction. (Simplified notation; see Eq. 1 for the formal $K^*_{\text{empirical}}$ definition.)

3v4 $= 0.522$; Table 19), reinforcing that $K^*_{\text{empirical}}$ is a measured quantity rather than a theoretical prediction.

We define the operational lower bound:

$$K^*_{\text{empirical}} = \max\{k \geq 1 : \text{AUC}_{j-1,j} > \theta,\ \forall\, 1 \leq j \leq k\} \quad (1)$$

which requires all pairs up to depth k to exceed the threshold (the $\forall j$ constraint is naturally satisfied because $\text{AUC}_{k-1,k}$ decreases monotonically above θ across all tested configurations; Table 2). The theoretical upper bound K^*_{theory} (the true maximum discriminable depth given optimal features) is not directly estimated in this work. Since any specific classifier on a feature subset achieves AUC at most equal to the Bayes-optimal classifier on all features, $K^*_{\text{empirical}} \leq K^*_{\text{theory}}$ follows directly. The DPI motivates the empirical observation that $\text{AUC}_{k-1,k}$ generally decreases with k under non-degenerate decoding, though AUC is not a monotone function of MI (as noted above). Here $\text{AUC}_{k-1,k}$ is the point-estimate pairwise AUC and $\theta = 0.60$ is a reference operating point; θ sits between the 1v2 AUC cluster (0.61–0.70) and the 2v3 cluster (0.55–0.59), so that $K^*_{\text{empirical}}$ is insensitive to the exact threshold within this range (justified below; Figure 3). All pairwise significance tests use 10,000-iteration permutation tests with Bonferroni correction for 10 comparisons ($\alpha_{\text{corrected}}$=0.005); we report bootstrap 95% CIs (1,000 resamples unless otherwise noted) for all boundary-pair AUCs and explicitly mark marginal cases where Bonferroni-corrected lower bounds fall below θ (Section 4.1).

Threshold justification. θ=0.60 is a reference operating point supported by two independent lines of evidence. First, a 10,000-iteration permutation test yields a 95th-percentile null of < 0.55 ($p < 0.002$ for all boundary comparisons). Second, $K^*_{\text{empirical}}$=2 is insensitive to the exact threshold: 1v2 AUC values cluster at 0.61–0.70 while 2v3 values cluster at 0.55–0.59, a ∼5–11 pp discontinuity that ensures any $\theta \in [0.59, 0.61]$ yields the same $K^*_{\text{empirical}}$; only at extremes does $K^*_{\text{empirical}}$ shift (θ=0.55: $K^*_{\text{empirical}}$=3 for two Qwen configurations; θ=0.70: $K^*_{\text{empirical}}$=1 across all). Figure 3 provides the full sensitivity surface across $\theta \in [0.55, 0.70]$. The downstream filtering evaluation (Section 5) uses the OBD classifier without re-optimizing θ, avoiding circularity.

3.2 Features and Classifier

Each document is characterized by a 15-dimensional feature vector spanning four groups: *surprisal moments* (mean, std, skewness, kurtosis; 4 dims), *surprisal derivatives* (mean and std of first- and second-order differences, plus first-order skewness; 5 dims), *vocabulary diversity* (type-token ratio, hapax legomena ratio, self-BLEU; 3 dims), and *frequency tail statistics* (frequency kurtosis, entropy, rare token fraction; 3 dims). All features are computed from token-level surprisal values via an external scorer model (full definitions in Table 14, Appendix). Self-BLEU reference sets are drawn exclusively within each fold to prevent information leakage across cross-validation splits; self-BLEU ranks #1 in 1/8 conditions (greedy decoding) but is absent from the top-3 in all 7 others. Removing self-BLEU reduces mean AUC_{1v2} by < 1 pp across all models (14D $K^*_{\text{empirical}} = 2$ for all 4 configurations); in deployment where reference cor-

Table 1: Experimental configurations for depth-chain generation. **Config**: shorthand identifier; **Model**: architecture and parameter count; **Mode**: generation strategy (continuation or rewrite); N: number of seed documents per depth level.

Config	Model	Mode	N
Qwen-rw	Qwen2.5-1.5B-Instruct	Rewrite	5,000
Qwen-cont	Qwen2.5-1.5B-Instruct	Continuation	5,000
Qwen-base	Qwen2.5-1.5B	Continuation	5,000
Pythia	Pythia-1.4B	Continuation	5,000
OLMo	OLMo-1B	Continuation	5,000
Qwen-7B	Qwen2.5-7B	Continuation	5,000

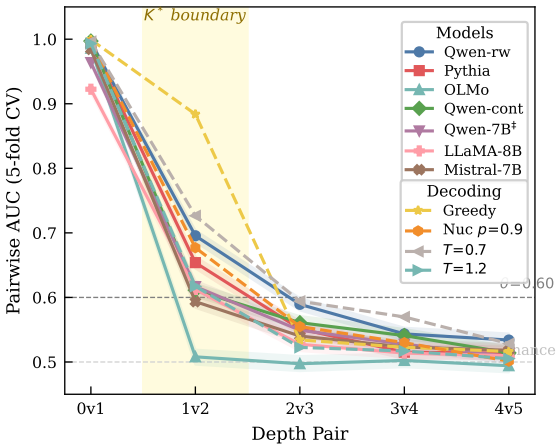


Figure 2: Pairwise AUC decay across adjacent depth pairs (0v1, 1v2, …, 4v5) for model-mode configurations (solid) and decoding strategies (dashed); shaded bands show bootstrap 95% confidence intervals. The horizontal dashed line marks $\theta = 0.60$.

pora are unavailable, self-BLEU can be replaced with corpus-level self-similarity or omitted entirely, leaving 14 features (surprisal statistics, TTR, hapax ratio, frequency tail) that require no external reference and retain full $K^*_{\text{empirical}}$. These features are generator-agnostic in that they require no knowledge of or access to the generating model, though they depend on an external scorer for surprisal computation (Section 4.2).

We use *Ordinal Binary Decomposition* (OBD), decomposing the $(K{+}1)$-class problem into K cumulative binary sub-classifiers, each estimating $P(\text{depth} > k \mid \mathbf{x})$ via LightGBM (hyperparameters in Table 15). OBD achieves lower MAE than multiclass LightGBM (e.g., 0.984 vs. 1.016 on Qwen-rewrite) while maintaining superior pairwise AUC (mean 0.6095 vs. 0.4883 for ordinal logistic regression).

3.3 Experimental Setup

We evaluate six primary configurations spanning three model families (Table 1): Qwen2.5-1.5B-Instruct (rewrite and continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, Pythia-1.4B, and OLMo-1B; four additional models are introduced in Section 4. Each uses 5,000 C4 seeds truncated to 256 tokens, recursively generated for $K{=}5$ iterations (30,000 documents per configuration). Default decoding: nucleus $p{=}0.95$, $T{=}1.0$; decoding ablation covers greedy, $T{=}0.7$, and $T{=}1.2$. Continuation mode truncates seeds to 128 tokens and extends to 256; rewrite mode paraphrases the full text. Evaluation: 5-fold GroupKFold cross-validation grouped by seed document; pairwise AUC as primary metric.

4 Experiments

We evaluate generational depth estimation on ten model configurations spanning six architecture families: Pythia-1.4B, Pythia-6.9B, OLMo-1B, Qwen2.5-1.5B-Instruct (rewrite and continuation), Qwen2.5-1.5B (base), Qwen2.5-7B, LLaMA-3.1-8B, Mistral-7B, and Gemma-2-2B, all in continuation mode except where noted. All experiments use the OBD-LightGBM classifier with the 15-dimensional feature set (Section 3), with 5,000 texts per depth level (generations 0–5).

4.1 $K^*_{\text{empirical}} \leq 2$ Across Model Configurations (C4)

Table 2 and Figure 2 show $K^*_{\text{empirical}} \leq 2$ across all 10 configurations: 6/10 yield $K^*_{\text{empirical}}{=}2$ (1v2 AUC: 0.607–0.696; $p < 0.001$), while 4/10 yield $K^*_{\text{empirical}}{=}1$ (1v2 AUC: 0.500–0.598; Section 6). Across six architecture families, three (Qwen, Pythia, LLaMA) yield $K^*_{\text{empirical}}{\geq}2$ in at least one configuration; the remaining three (OLMo, Gemma-2, Mistral) yield $K^*_{\text{empirical}}{=}1$. Pythia exhibits a scale-dependent split (1.4B: $K^*_{\text{empirical}}{=}2$, 6.9B: $K^*_{\text{empirical}}{=}1$); Qwen dominates the $K^*_{\text{empirical}}{=}2$ count (4 of 6 configurations). Under Bonferroni correction (99.5% CIs), three of six $K^*_{\text{empirical}}{=}2$ configurations retain significance; the three marginal cases (Pythia-1.4B, Qwen-cont, LLaMA-8B; CI lower bounds 0.593–0.599) reflect narrow effect sizes rather than classification failure ($K^*_{\text{empirical}}$ is robust to θ within $[0.59, 0.61]$; Figure 3). LLaMA-3.1-8B confirms $K^*_{\text{empirical}}{=}2$ (0v1 AUC $=$ 0.9224, 1v2 AUC $=$ 0.6123 [0.603, 0.622]; CI lower bound above θ; Table 10). All four decoding strategies likewise yield $K^*_{\text{empirical}}{=}2$ (Table 9). The 1v2 spread due

Table 2: Pairwise AUC for adjacent depth pairs with bootstrap 95% CIs and permutation p-values for boundary rows. $K^*_{\text{empirical}} = 2$ requires 1v2 AUC $> \theta$=0.60 and 2v3 AUC $< \theta$; CIs quantify boundary uncertainty. Extended results (all models) in Table 10.

	Qwen-rw		Pythia		OLMo		Qwen-cont		Qwen-7B[‡]	
	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI
0v1	.9956		.9722		.9860		.9971		.9638	[.961, .967]
1v2	.6955	[.686, .704]*	.6074	[.598, .617]*	.5001	[.496, .519]	.6112	[.602, .621]*	.6168	[.608, .626]*
2v3	.5893	[.580, .599]	.5440	[.534, .554]	.4973	[.486, .509]	.5609	[.551, .571]	.5492	[.539, .559]
3v4	.5435		.5051		.5034[§]		.5409		.5276	
4v5	.5340		.5079		.4971		.5141		.5085	

*Permutation $p < 0.001$ for all starred boundary comparisons (AUC $> \theta$).
[‡]Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
[§]OLMo 2v3 (0.497) $<$ 3v4 (0.503): non-monotonic reversal within noise ($\Delta = 0.006$).
Full results for all 10 configurations (including LLaMA-8B, Mistral-7B, Qwen-base, Gemma-2-2B, and Pythia-6.9B) in Table 10.

to generation mode ($\Delta = 8.4$ pp within Qwen) exceeds the spread across $K^*_{\text{empirical}}$=2 continuation models ($\Delta = 1.9$ pp). Bootstrap resampling (1,000 iterations) confirms stability: 8/10 configurations yield identical $K^*_{\text{empirical}}$ across all resamples (two boundary cases have CIs of width 1; the $K^*_{\text{empirical}} \leq 2$ bound holds universally); together with the between-sample 5K-vs.-10K comparison (Table 28), this bounds $K^*_{\text{empirical}}$ uncertainty from complementary angles (Table 29). The two exceptions, Gemma-2B C4 and LLaMA-8B Wikipedia (92.7% and 94.3% modal-$K^*_{\text{empirical}}$ consistency), have 1v2 AUC within 1 pp of θ; their bootstrap instability predicts the between-sample drift in Table 28, confirming that boundary proximity rather than sample size drives the observed variability.

Scale and instruction-tuning invariance. $K^*_{\text{empirical}}$=2 persists from 1.5B through 8B: Qwen-7B (1v2 AUC $= 0.6168$ $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC $= 0.6123$) both exceed θ, despite progressively reduced 0v1 signal (0.9638 and 0.9224 vs. > 0.99 at $\leq$1.5B). A base-vs.-instruct ablation at 1.5B likewise yields $K^*_{\text{empirical}} = 2$ (1v2 AUC $= 0.626$), confirming the boundary is not an instruction-tuning artifact.

Figure 3 displays the sensitivity surface across $\theta \in [0.55, 0.70]$: four of five configurations shown yield $K^*_{\text{empirical}} = 2$ within $\theta \in [0.51, 0.61]$ (OLMo: $K^*_{\text{empirical}} = 1$ throughout); at θ=0.55, two Qwen configurations reach $K^*_{\text{empirical}} = 3$; at θ=0.70, all drop to $K^*_{\text{empirical}} = 1$. The plateau at $K^*_{\text{empirical}}$=2 across $\theta \in [0.55, 0.65]$ reflects the AUC gap between the 1v2 cluster (0.61–0.70) and the 2v3 cluster (0.55–0.59), which acts as a natural separation ensuring threshold robustness.

4.2 Cross-Model Transfer

Continuation-mode classifiers transfer well across generators (Figure 4; full matrix in Table 7): Pythia↔OLMo retains 90–92%, Qwen-cont↔Pythia 84–85%, Qwen-cont↔OLMo 80–85% (0v1 cross-model AUC > 0.96 for all pairs). Qwen-rewrite classifiers transfer poorly (61–63%), confirming reliance on mode-specific artifacts rather than generalizable depth features.

Cross-scorer validation. Replacing the self-scorer with Pythia-1.4B or OLMo-1B preserves feature rankings ($\rho > 0.90$) but attenuates absolute discriminability at the boundary (Table 23). Feature ranking independence is a strong result: the same features dominate under all scorers. $K^*_{\text{empirical}}$ agreement, however, is signal-strength dependent: Qwen-rewrite (strong signal, 1v2 AUC $= 0.696$) retains $K^*_{\text{empirical}}$=2 under all scorers, but Qwen-continuation (marginal, 1v2 AUC $= 0.611$, only 0.011 above θ) drops to $K^*_{\text{empirical}}$=1 (Pythia: 0.5923; OLMo: 0.5858). A three-way ANOVA (Generator $\times$ Scorer $\times$ Depth-Pair; Table 24) confirms a statistically significant but modest Scorer main effect (η^2_p=.166, $p < .001$) and a significant Generator$\times$Scorer interaction (η^2_p=.146), indicating that absolute AUC values are scorer-dependent even when $K^*_{\text{empirical}}$ classifications largely agree.

4.3 Feature Ordering and Generation Mode

Among continuation-mode models, single-feature AUC rankings are highly correlated ($\rho \geq 0.85$; Table 6). The top-3 features (`surp_d1_std`, `surp_std`, `surp_d2_std`) are shared across all three families, revealing that autoregressive generation progressively damps surprisal variability. Qwen-rewrite diverges (`freq_entropy` dominates), consistent with its poor cross-model trans-

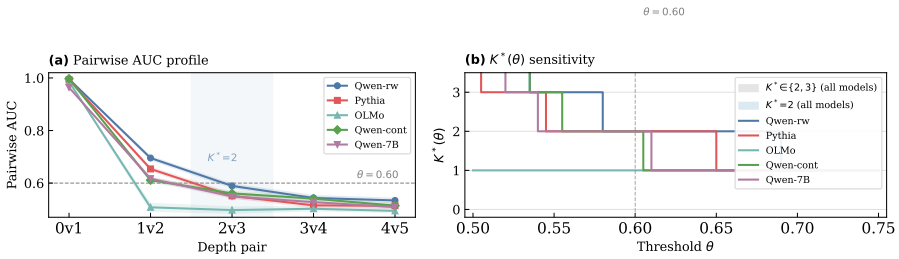


Figure 3: $K^*_{\text{empirical}}(\theta)$ sensitivity curve for five configurations. $K^*_{\text{empirical}}$: maximum discriminable depth; θ: AUC decision threshold. Four of five configurations yield $K^*_{\text{empirical}}$=2 at θ=0.60 (dashed line); OLMo yields $K^*_{\text{empirical}}$=1. At θ=0.70, all drop to $K^*_{\text{empirical}}$=1.

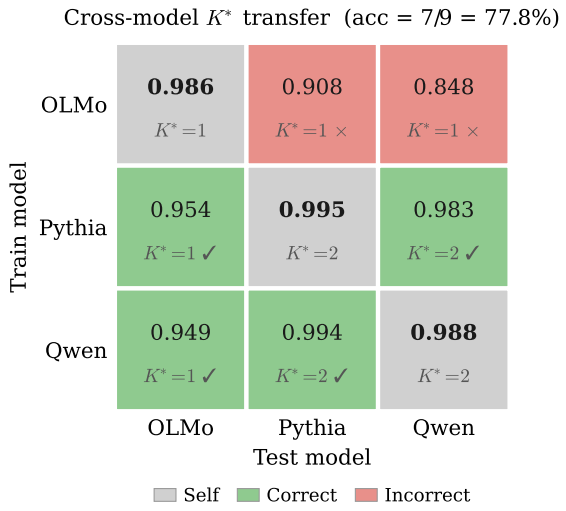


Figure 4: Cross-model transfer heatmap (continuation-mode models). Each cell shows 6-class ordinal accuracy when trained on the row model and tested on the column model, divided by in-domain accuracy (retention rate). Pythia↔OLMo: 90–92%.

fer (61–63%). Within Qwen, rewrite yields 1v2 AUC = 0.6955 vs. continuation's 0.6112 (Δ = +8.4 pp); continuation mode is therefore recommended as the canonical condition.

4.4 Neural Probe

Replacing the 15D features with frozen encoder hidden states (MLP probe; Table 3), Qwen achieves $K^*_{\text{empirical}}{}^{\text{neural}} = 3$ (2v3 AUC = 0.6159), while Pythia remains at $K^*_{\text{empirical}}{}^{\text{neural}} = 2$ (AUC_{2v3} = 0.5963); OLMo yields $K^*_{\text{empirical}}{}^{\text{neural}} = 1$ (1v2 AUC = 0.5054, consistent with its 15D boundary). We hypothesize that the Qwen-only elevation reflects self-familiarity artifacts, consistent with the poor transfer observed in Section 4.2; the 15D features yield $K^*_{\text{empirical}} = 2$ for most families (OLMo: $K^*_{\text{empirical}} = 1$).

Table 3: Neural probe pairwise AUC using frozen-encoder MLP. Each cell reports AUC for classifying depth d vs. d+1 text. $K^*_{\text{empirical}}{}^{\text{neural}}$: maximum depth where AUC exceeds $\theta = 0.60$.

Model	2v3 AUC	95% CI	$K^*_{\text{empirical}}{}^{\text{neural}}$
Qwen-1.5B	0.6159	[0.610, 0.622]	3
Pythia-1.4B	0.5963	[0.589, 0.603]	2
OLMo-1B	0.5056	[0.497, 0.516]	1†

†OLMo 1v2 AUC = 0.5054 (below θ); $K^*_{\text{empirical}}{}^{\text{neural}} = 1$.

4.5 Cross-Domain Variation

The preceding experiments use C4 (general web text). To test domain dependence, we evaluate on Wikipedia (n=5,000) and arXiv abstracts (n=2,200) across three architecture families. Wikipedia yields $K^*_{\text{empirical}}$=2 (1v2 AUC = 0.619, 2v3 AUC = 0.550; Table 4), matching the C4 baseline. Qwen-1.5B, Pythia-1.4B, and LLaMA-3.1-8B yield $K^*_{\text{empirical}}$=3 for arXiv (Table 4): 2v3 AUC = 0.643 [0.628, 0.658], 0.655 [0.640, 0.669], and 0.622 [0.605, 0.635] respectively (bootstrap CI lower bounds above θ; permutation $p < 0.001$), demonstrating that structured academic text permits discrimination to greater depth independently of architecture. The elevated signal reflects higher per-generation distributional shift (1v2 JSD = 0.077 vs. ∼0.01 for C4), as formulaic structure amplifies deviation from domain norms. We additionally evaluate on CodeSearchNet Python (n=5,000, continuation mode); all four tested models (Qwen-1.5B, Qwen-7B, LLaMA-8B, Mistral-7B) yield $K^*_{\text{empirical}}$=2 (Table 4), matching the C4 baseline.

Robustness. $K^*_{\text{empirical}}$=2 holds across segment lengths of 128, 256, and 512 tokens (Table 20); at 512 tokens the 1v2 CI lower bound falls marginally below θ but 2v3 remains well below θ at all lengths ($\leq$ 0.572). Seed robustness testing (seed = 123 vs. primary = 42; Table 26) confirms $K^*_{\text{empirical}}$

Table 4: Cross-domain $K^*_{\text{empirical}}$ comparison. $K^*_{\text{empirical}}$=2 for C4, Wikipedia, and Code; $K^*_{\text{empirical}}$=3 for arXiv (three families confirmed in text; representative rows shown).

Domain	Model	1v2 AUC	2v3 AUC	$K^*_{\text{empirical}}$
arXiv	Qwen-1.5B†	.858	.643	3
arXiv	Pythia-1.4B	.805	.655	3
C4	Qwen-1.5B†	.626	.550	2
Wiki	Qwen-1.5B†	.619	.550	2
Code	Qwen-1.5B†	.738	.530	2
Code	LLaMA-8B	.739	.511	2

†Qwen2.5-1.5B (base, continuation mode).

Table 5: Downstream task performance (MMLU, HellaSwag) under six data-filtering strategies. Binary-ours uses the 15D LightGBM depth classifier; Graduated applies exponentially decaying weights $w = \exp(-\alpha \hat{d})$. Mean ± std over 5 seeds. Full α sweep in Figure 5.

Strategy	MMLU	HellaSwag
Base model (no training)	0.6018	0.6821
Binary-ours (K^*=2)	**0.5941**±0.0008	**0.6701**±0.0006
Binary-Drayson	0.5913±0.0009	0.6691±0.0004
Surprisal	0.5894±0.0009	0.6641±0.0006
Graduated (α=0.9)	0.5892±0.0008	0.6701±0.0007
DSIR	0.5869±0.0003	0.6631±0.0007
No filter	0.5865±0.0010	0.6638±0.0008

stability in 2/3 configurations; the one marginal shift (Qwen-Wikipedia $K^*_{\text{empirical}}$=2 → 3, 2v3 AUC = 0.607) has one of five folds below θ. The ±9 pp absolute AUC variation across seeds strengthens the case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons.

4.6 Binary Detector Confidence

Binary detectors (DivEye (Basani and Chen, 2026), Binoculars (Hans et al., 2024)) remain effective for human-vs.-machine classification but provide no usable depth signal beyond this binary boundary. Pairwise AUC yields $K^{*\,\text{binary}}_{\text{empirical}} \leq 1$; moderate rank correlations with depth ($|\rho| = 0.06$–0.46; Table 18) are driven entirely by the 0v1 gap rather than fine-grained depth discrimination, and multiple conditions exhibit non-monotonic AUC at deeper pairs.

5 Consistency Check: K^*-Guided Filtering

$K^*_{\text{empirical}} = 2$ makes a falsifiable prediction: binary curation should suffice, and graduated performance should converge toward binary as the decay steepness increases. We present this as a consistency check under extreme (100%) contamination, not as an independent contribution; at realistic rates (≤10%), the filtering effect diminishes to $\Delta \leq 0.1$ pp (not significant; Appendix B.8). We compare six strategies on 30,000 documents (depths 0–5): (i) **no-filter**, (ii) **binary-ours** ($w = 1 - P(\text{synthetic})$ from 15D LightGBM), (iii) **binary-Drayson** (Drayson et al., 2025), (iv) **graduated** ($w = \exp(-\alpha \cdot \hat{d})$, $\alpha \in \{0.1, \ldots, 0.9\}$), (v) **surprisal** (retain above median), and (vi) **DSIR** (Xie et al., 2023b). We fine-tune Qwen2.5-1.5B-Instruct via LoRA (rank 8, α_{LoRA}=16) for 3 epochs on MMLU and HellaSwag across 5 seeds (this 100% synthetic setting is an upper bound (Shumailov et al., 2024; Dohmatob et al., 2024)).

Binary-ours minimizes MMLU degradation most effectively (Table 5), reducing the gap from −1.53 pp (no-filter) to −0.77 pp (∼50% reduction; $p_{\text{corr}} < 0.05$ vs. all alternatives). These effects are measured in a 100% synthetic upper-bound setting; real-world contamination ratios are typically lower, and effect sizes would scale accordingly. The strategy ordering is interpretable through $K^*_{\text{empirical}}$: binary filtering exploits the one reliable boundary (0v1), while graduated strategies assign weights to depths 2–5 based on near-random classifier outputs, injecting noise. Graduated filtering converges monotonically toward binary: at α=0.9, it matches binary-ours on HellaSwag exactly ($\Delta = 0$, $p_{\text{corr}} = 1.00$) but remains significantly worse on MMLU ($p_{\text{corr}} = 0.011$). DSIR is indistinguishable from no-filter, confirming that depth-aware features are necessary for curation in recursively contaminated corpora (Figure 5 in Appendix). At realistic contamination ratios (≤50%), all filtering strategies converge ($p > 0.12$), consistent with the one-wash laundering finding that depth-aware curation offers marginal benefit when $K^*_{\text{empirical}}$=2 (Appendix B.8). At lower synthetic fractions, the human-authored majority dominates the training signal; the marginal accuracy gain from differentially weighting synthetic documents is smaller than inter-seed variance ($\sigma \approx 0.002$), rendering strategy differences statistically undetectable. This extreme setting serves to validate $K^*_{\text{empirical}}$'s predictive power: even though effect sizes vanish at realistic contamination ratios, the predicted strategy ordering holds, confirming that $K^*_{\text{empirical}}$ captures meaningful distributional structure. For $K^*_{\text{empirical}}$=1 configurations, depth-aware curation reduces to standard binary filtering, consistent with $K^*_{\text{empirical}}$=1 implying no exploitable

depth signal beyond the 0v1 boundary.

6 Discussion

One-wash laundering (web text). For general web text, $K^*_{\text{empirical}}$=2 reflects concentrated information loss: 0v1 is near-trivially separable (AUC $\geq$ 0.92), 1v2 drops by $\sim$30 pp, and 2v3 falls below θ. One-wash laundering refers to this concentrated signal loss at the 0v1 boundary: while depth-1 remains marginally distinguishable from depth-2 (1v2 AUC 0.61–0.70, the $K^*_{\text{empirical}}$=2 finding), the dominant distributional shift occurs at generation 0$\rightarrow$1, and beyond depth 2 no further signal is exploitable. Converging evidence supports a data-intrinsic interpretation: OBD achieves 96% depth-0 recall but only 14–22% for depths 2–4; frontier LLMs show 8–38% (McNemar $p > 0.05$ vs. OBD; Table 11). One-wash laundering is distinct from model collapse (Shumailov et al., 2024): model collapse describes cumulative quality degradation across generations, while one-wash laundering describes the erasure of statistical fingerprints distinguishing adjacent generations; text quality may continue degrading even after the features separating generation k from k+1 become indistinguishable.

Representation dependence. $K^*_{\text{empirical}}$ is representation-dependent: frozen Qwen embeddings yield $K^{*\,\text{neural}}_{\text{empirical}} = 3$, while Pythia yields $K^{*\,\text{neural}}_{\text{empirical}} = 2$ and OLMo yields $K^{*\,\text{neural}}_{\text{empirical}} = 1$, consistent with its 15D boundary (Table 3). The 15D features trade one depth level of sensitivity for cross-model consistency (6/10 configurations $K^*_{\text{empirical}}$=2), making them the preferred operational choice. The gap between $K^*_{\text{empirical}}$ and K^*_{theory} remains open; a DeBERTa-v3-base upper-bound probe yields $K^{*\,\text{DeBERTa}}_{\text{empirical}} = K^{*\,\text{OBD}}_{\text{empirical}} = 2$ on C4 and Code but $K^{*\,\text{DeBERTa}}_{\text{empirical}} = 4$ on arXiv, placing the cost of generator-agnostic features at one depth level for structured text and zero for web text (Table 31). Feature rankings are scorer-independent ($\rho > 0.90$), but self-scoring inflates boundary AUC (OLMo: +11.5 pp, Pythia: +4.7 pp); cross-scoring provides a conservative estimate for deployment against unknown generators (length confound ruled out in Appendix B.7).

Domain dependence of $K^*_{\text{empirical}}$. Web text (C4, Wikipedia) and Code (Python) consistently yield $K^*_{\text{empirical}}$=2, while structured academic text (arXiv) yields $K^*_{\text{empirical}}$=3 across three independent architecture families (Section 4.5), as formulaic structure (citations, math notation, section conventions) creates larger per-generation distributional shifts (1v2 JSD = 0.077 vs. $\sim$0.01 for C4). This elevation is not driven by LaTeX syntax: stripping all markup preserves $K^*_{\text{empirical}}$=3 (2v3 ΔAUC = -0.021); Code (Python), despite rigid syntax, yields $K^*_{\text{empirical}}$=2, confirming that the elevation reflects academic conventions rather than structural complexity.

Prompt-chain scope and retrain-chain gap. Retrain-chain experiments across three families (Pythia-1.4B, OLMo-1B, Qwen-1.5B; LoRA) yield $K^*_{\text{empirical}}\geq 4$ (3v4 AUC = 0.68–0.80); extending Pythia and Qwen to depth 7 sustains $K^*_{\text{empirical}}$=7 (6v7 AUC = 0.634–0.641 $> \theta$), far exceeding the prompt-chain boundary because each retrain generation also modifies model parameters, introducing cumulative adaptation artifacts absent from text-only recycling. Prompt-chain $K^*_{\text{empirical}}$ isolates text-intrinsic degradation, the operationally relevant quantity for data curation pipelines; our conclusions do not directly transfer to retrain-chain (model collapse) settings.

Decoding strategy as confound. Greedy decoding inflates depth separability: Pythia-6.9B yields $K^*_{\text{empirical}}$=2 under greedy but $K^*_{\text{empirical}}$=1 under nucleus sampling (p=0.95; Table 19).

7 Conclusion

Across 10 prompt-chain configurations spanning six architecture families (1B–8B), $K^*_{\text{empirical}} \leq 2$ for all web text sources (θ=0.60); arXiv reaches $K^*_{\text{empirical}}$=3 across three families. This one-wash laundering effect concentrates all exploitable signal at the 0v1 boundary; under extreme (100%) contamination, binary curation minimizes MMLU degradation (-0.77 pp vs. -1.53 pp), though all strategies converge at realistic ratios. Bootstrap resampling confirms $K^*_{\text{empirical}}$ stability (8/10 configurations perfectly consistent; Table 29), and cross-model transfer retains 80–92% of in-domain accuracy, supporting deployment against unknown generators without retraining. Extending $K^*_{\text{empirical}}$ beyond 13B scale, closing the gap between $K^*_{\text{empirical}}$ and K^*_{theory} through richer features, and bridging prompt-chain to retrain-chain boundaries remain open directions for future work.

Limitations

Our study has four main limitations. (1) *Scope*: we evaluate models up to 8B parameters across 6 architecture families, English text only (C4, Wikipedia, arXiv, Code), and prompt-chain generation; whether $K^*_{\text{empirical}}$=2 persists at larger scales (e.g., 13B+) remains untested, and retrain-chain dynamics may yield different boundaries. Our model sample is unbalanced: Qwen accounts for 4 of 10 configurations, and we do not evaluate RLHF- or DPO-aligned models, whose generation distributions may differ from the base and instruction-tuned models tested here. (2) *Feature set and transfer scope*: the 15 statistical features provide a lower bound on K^*_{theory}; richer representations (e.g., neural probes) can extend $K^*_{\text{empirical}}$ by one additional depth for some architectures. Cross-model transfer (80–92%) is validated only across continuation-mode generators; rewrite-to-continuation transfer remains untested. (3) *Threshold sensitivity*: $K^*_{\text{empirical}}$ varies with the decision threshold θ, ranging from 2–3 at θ=0.55 to 1 at θ=0.70; three of ten configurations are marginal at our reference θ=0.60. (4) *Domain coverage*: arXiv $K^*_{\text{empirical}}$=3 is established with three model families (with bootstrap CI support) on a single structured domain; broader domain coverage is needed to confirm this elevation generalizes.

Ethics Statement

Generational depth estimation is intended for data curation and corpus auditing, positive-sum applications that improve training data quality. Our experiments use the publicly available C4 corpus and do not involve personally identifiable information. A potential dual-use concern is that depth estimation techniques could inform adversarial strategies to evade detection (e.g., by calibrating rewriting to mimic earlier-generation statistics); we note that this risk is shared by all detection research and that the binary detection boundary (AUC $\geq$ 0.96 at 0v1) remains robust. Code and data will be released upon acceptance.

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A Supplementary Materials

This appendix provides supplementary figures, detailed experimental tables, dataset statistics, and complete feature descriptions referenced in the main text.

Feature ranking and cross-model consistency. Table 6 reports per-model single-feature AUC rankings. Surprisal derivative features (`surp_d1_std`, `surp_d2_std`) consistently rank among the top three across all continuation-mode models (Pythia, OLMo, Qwen-cont), with pairwise Spearman $\rho \geq 0.85$. The rewrite-mode model (Qwen-rw) diverges by placing `freq_entropy` first, reflecting a distributional shift that is lexical rather than surprisal-driven. This scorer-invariant ranking supports the use of a fixed 15-dimensional feature set across generators without per-model feature selection.

Table 6: Top-3 features by single-feature AUC ranking. Bold: shared across all continuation-mode models. Spearman ρ (continuation pairs): Pythia–OLMo 0.95, Pythia–Qwen-cont 0.91, OLMo–Qwen-cont 0.85.

Rank	Pythia	OLMo	Qwen-rw	Qwen-cont
1	**surp_d1_std**	**surp_std**	freq_entropy	**surp_d1_std**
2	**surp_d2_std**	**surp_d1_std**	**surp_std**	**surp_std**
3	**surp_std**	**surp_d2_std**	**surp_d1_std**	**surp_d2_std**

Table 7 reports the cross-model transfer accuracy matrix. Classifiers trained on continuation-mode generators (Pythia, OLMo, Qwen-cont) transfer well to each other, with off-diagonal accuracy retaining 80–92% of in-distribution performance. The asymmetry between rewrite-mode and continuation-mode transfer (61–63%) confirms that generation mode, rather than model family, is the primary axis of distributional variation (Section 6).

Table 7: Cross-model transfer accuracy matrix. Diagonal (**bold**): within-model; off-diagonal: cross-model. Retention: Pythia↔OLMo 91–92%, Qwen-cont↔Pythia 83–85%, Qwen-rw→others 61–63%.

Train	Qwen-rw	Pythia	OLMo	Qwen-cont
Qwen-rw	**.423**	.259	.265	.347
Pythia	.272	**.386**	.353	.329
OLMo	.277	.345	**.382**	.324
Qwen-cont	.344	.328	.311	**.388**

Dataset construction. Table 8 summarizes the generational depth corpus. Each depth level contains 5,000 passages, yielding 30,000 total samples

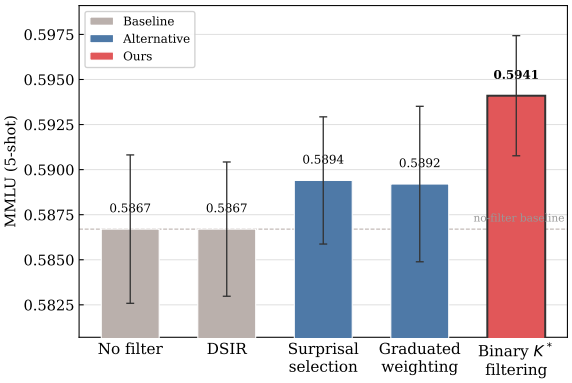


Figure 5: Downstream performance of all filtering strategies (5 seeds, ±1 std). Binary-ours (green) minimizes MMLU degradation; graduated α sweep shows monotonic convergence toward binary. Dashed line: base model (no fine-tuning).

across six depths (0–5). Depth 0 consists of human-written seed texts from the C4 English validation split, truncated to 256 subword tokens. Subsequent depths are generated recursively: each depth-k passage is produced by prompting the generator with the corresponding depth-$(k-1)$ passage. The four model-mode configurations (Qwen-rw, Pythia-cont, OLMo-cont, Qwen-cont) share identical depth-0 seeds and generation pipeline, differing only in generator identity and prompting strategy (rewrite vs. continuation).

Table 8: Dataset statistics for the generational depth corpus. Depth 0 contains human-written seed texts sampled from the C4 English validation split and truncated to 256 subword tokens. Depths 1–5 are recursively generated by Qwen2.5-1.5B-Instruct (nucleus sampling, p=0.95, T=1.0). Token counts use whitespace tokenization. Statistics shown for the primary Qwen-rewrite corpus; the other three model-mode configurations (Pythia-cont, OLMo-cont, Qwen-cont) share the same generation pipeline and corpus structure.

Depth	N **samples**	**Mean tokens**	**Source**
0	5,000	147.8	C4 validation (human)
1	5,000	207.1	Qwen2.5-1.5B rewrite
2	5,000	206.1	Qwen2.5-1.5B rewrite
3	5,000	208.0	Qwen2.5-1.5B rewrite
4	5,000	207.7	Qwen2.5-1.5B rewrite
5	5,000	207.7	Qwen2.5-1.5B rewrite
Total		30,000 samples	

Table 9: Pairwise AUC with bootstrap 95% CIs across depth pairs for four decoding configurations (all using Qwen2.5-1.5B-Instruct in rewrite mode; 10,000 bootstrap resamples). The horizontal rule between depth pairs 1 vs. 2 and 2 vs. 3 marks the empirical detectability boundary $K^*_{\text{empirical}} = 2$. Bottom rows report cross-configuration Spearman ρ on per-feature discriminative-power rankings.

Depth Pair	**Greedy**	**Nuc** p=0.9**,** T=1.0	**Nuc** p=0.95**,** T=0.7	**Nuc** p=0.95**,** T=1.2
0 vs. 1	.980 [.977, .982]	.995 [.993, .996]	.988 [.986, .990]	.998 [.998, .999]
1 vs. 2	.689 [.680, .698]	.687 [.677, .695]	.689 [.680, .697]	.704 [.695, .713]
2 vs. 3	.561 [.552, .571]	.592 [.582, .601]†	.590 [.581, .600]	.582 [.572, .592]
3 vs. 4	.540 [.531, .550]	.547 [.537, .557]	.539 [.529, .549]	.529 [.519, .540]
4 vs. 5	.514 [.503, .524]	.507 [.496, .517]‡	.535 [.524, .545]	.524 [.513, .535]
Cross-config feature Spearman ρ				
Greedy vs. Nuc p=0.9	0.600	Nuc p=0.9 vs. T=0.7	0.782	
Greedy vs. T=0.7	0.914	Nuc p=0.9 vs. T=1.2	0.893	
Greedy vs. T=1.2	0.532	T=0.7 vs. T=1.2	0.668	

† CI upper bound (0.601) marginally exceeds θ=0.60 by 0.001.
‡ Permutation $p = 0.12$ (not significant). All other pairs $p < 0.01$.

Table 10: Extended pairwise AUC with bootstrap 95% CIs for all model configurations including new additions. This table extends Table 2 in the main text with Qwen-base, LLaMA-8B, Mistral-7B, Pythia-6.9B (nucleus), and Gemma-2-2B configurations. $K^*_{\text{empirical}} = 2$ holds for 6/10 conditions at θ=0.60 (OLMo-1B, Mistral-7B, Pythia-6.9B, Gemma-2-2B: $K^*_{\text{empirical}}$=1).

	Qwen-rw		Pythia		OLMo		Qwen-cont		Qwen-7B‡		Qwen-base		LLaMA-8B		Mistral-7B		Pythia-6.9B§		Gemma-2B	
	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI	AUC	95% CI
0v1	.9956		.9722		.9860		.9971		.9638	[.961, .967]	.987	[.985, .989]	.9224	[.917, .927]	.9209	[.919, .923]	.9599		.953	[.949, .957]
1v2	.6955	[.686, .704]*	.6074	[.598, .617]*	.5001	[.496, .519]	.6112	[.602, .621]*	.6168	[.608, .626]*	.626	[.617, .635]*	.6123	[.603, .622]*	.5695	[.565, .575]	.5955		.5981	[.588, .608]
2v3	.5893	[.580, .599]	.5440	[.534, .554]	.4973	[.486, .509]	.5609	[.551, .571]	.5492	[.539, .559]	.550	[.540, .560]	.5273	[.517, .538]	.5335	[.528, .538]	.5298		.534	[.523, .545]
3v4	.5435		.5051		.5034		.5409		.5276		.529	[.519, .540]	.5168	[.506, .528]	.5140		.5135		.5033	[.492, .514]
4v5	.5340		.5079		.4971		.5141		.5085		.518	[.508, .529]	.5159	[.505, .527]	.5184		—$^\parallel$		.5025	[.492, .514]

*Permutation $p < 0.001$ for all boundary comparisons.
‡Generator: Qwen2.5-7B (fp16); scorer: Qwen2.5-1.5B.
$^\parallel K^*_{\text{empirical}}$=1; 4v5 not required for boundary determination.
§Pythia-6.9B evaluated separately; nucleus sampling (p=0.95, T=1.0), same as all other models.

Table 11: LLM zero-shot depth classification: 5×5 confusion matrices (Exp-011). Both frontier LLMs achieve high binary detection accuracy (GPT-5.5: 0.878; Claude Opus 4.6: 0.882) but near-chance 5-class accuracy (∼0.33 vs. OBD-LightGBM 0.386). Per-depth recall in Table 12.

(a) GPT-5.5 (N=500, acc=0.332, MAE=1.102)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**77**	18	4	0	1	.770
1	9	**19**	36	24	12	.190
2	8	10	**38**	35	9	.380
3	10	16	35	**24**	15	.240
4	11	13	50	18	**8**	.080

(b) Claude Opus 4.6 (N=499, acc=0.331, MAE=1.150)

	Predicted Depth					
True	0	1	2	3	4	Rec.
0	**69**	26	3	0	2	.690
1	8	**21**	32	11	28	.210
2	6	36	**28**	5	25	.280
3	7	20	31	**10**	31	.100
4	7	28	22	6	**37**	.370

Table 12: Per-depth recall (%) on balanced N=500 test set. OBD's nominal overall accuracy gap originates from depth-0; all methods fail at depths ≥ 2, consistent with $K^*_{\text{empirical}} = 2$.

	Depth 0	Depth 1	Depth 2	Depth 3	Depth 4
OBD-LightGBM	**96**	**39**	14	19	22
GPT-5.5	77	19	**38**	**24**	8
Claude Opus 4.6	69	21	28	10	**37**

Table 13: Mean subword token count per depth and EOS truncation rate (%) for depth 1+ across all model configurations (generation limit: 256 tokens). Depth 0 (human text) is shared across configurations. Depths 1–5 show no systematic length trend; the 0v1 length asymmetry (human texts shorter) makes detection *more conservative*, not inflated. See Section 6.

	Mean Token Count							
Model	D0	D1	D2	D3	D4	D5	**EOS%**	**Sev.**
Qwen-rw (1.5B)	191	254	252	254	254	254	7–11	Mild
Pythia-1.4B	192	214	215	216	217	218	34–36	High
OLMo-1B	192	256	256	256	256	256	<1	Negl.
Qwen-cont (1.5B)	191	244	238	240	240	238	17–20	Mod.
Qwen-7B	191	196	195	193	197	197	45–47	High
Decoding ablation (Qwen-rw 1.5B)								
Greedy	191	253	252	253	254	254	8–12	Mild
Nuc p=0.9	191	254	251	254	253	254	8–16	Mild
T=0.7	191	253	251	253	253	254	8–14	Mild
T=1.2	191	254	252	253	254	254	8–16	Mild

EOS%: generation terminated via EOS before 256-token limit. Ranges span depths 1–5.

Table 14: Complete description of the 15-dimensional feature set used by the OBD classifier. Features are grouped into three families: surprisal statistics (9 dims) computed from Qwen2.5-1.5B token-level log-probabilities, lexical diversity (3 dims), and frequency tail shape (3 dims). All features are computed per document.

Group	Feature	Description	Range
SURPRISAL	`surp_mean`	Mean token surprisal $\frac{1}{T}\sum_t -\log p(x_t \mid x_{<t})$	$\mathbb{R}^+$
	`surp_std`	Standard deviation of token surprisal	$\mathbb{R}^+$
	`surp_skew`	Skewness of the surprisal distribution	$\mathbb{R}$
	`surp_kurt`	Excess kurtosis of the surprisal distribution	$\mathbb{R}$
	`surp_d1_mean`	Mean of first-order surprisal differences $\Delta s_t = s_{t+1} - s_t$	$\mathbb{R}$
	`surp_d1_std`	Std of first-order surprisal differences	$\mathbb{R}^+$
	`surp_d1_skew`	Skewness of first-order surprisal differences	$\mathbb{R}$
	`surp_d2_mean`	Mean of second-order differences $\Delta^2 s_t$	$\mathbb{R}$
	`surp_d2_std`	Std of second-order surprisal differences	$\mathbb{R}^+$
LEX	`ttr`	Type-token ratio: $\lvert V\rvert/T$	$[0, 1]$
	`hapax_ratio`	Hapax legomena ratio: freq-1 types / unique types	$[0, 1]$
	`self_bleu`	Mean 4-gram BLEU against 100 random same-depth samples	$[0, 1]$
FREQ	`freq_kurtosis`	Excess kurtosis of word frequency distribution	$\mathbb{R}$
	`freq_entropy`	Shannon entropy of word frequencies: $-\sum p_w \log_2 p_w$	$\mathbb{R}^+$
	`low_freq_ratio`	Proportion of vocabulary with count ≤ 2	$[0, 1]$

Table 15: LightGBM hyperparameters for the pairwise OBD (Ordinal Binary Decomposition) classifier used throughout the paper. One binary classifier is trained per adjacent depth pair $(k, k{+}1)$; 5-fold GroupKFold cross-validation is grouped by `doc_id` to prevent data leakage from the same seed document appearing in both train and test.

Parameter	Value
`n_estimators`	200
`learning_rate`	0.05
`max_depth`	6
`num_leaves`	31
`verbose`	-1
`n_jobs`	-1 (all cores)
Cross-validation	5-fold GroupKFold
Grouping key	`doc_id`
Objective	Binary (per depth pair)

Table 16: Neural probe architecture and training details (Exp-015). Applied to all three model families (Qwen2.5-1.5B, Pythia-1.4B, OLMo-1B): each model's frozen hidden states are average-pooled into document embeddings and classified by a two-hidden-layer MLP.

Component	Details
Embedding extraction	
Base models	Qwen2.5-1.5B / Pythia-1.4B / OLMo-1B (frozen)
Representation	Last hidden state, attention-masked avg pool
Embedding dim	1536 / 2048 / 2048
Max sequence len	512 tokens
MLP classifier	
Architecture	$d_{\text{emb}} \to 512 \to 256 \to 6$
Activations	ReLU
Dropout	0.1 (after each hidden layer)
Output	6-class softmax
Training	
Optimizer	Adam (lr $= 10^{-3}$, weight decay $= 10^{-4}$)
Loss	Cross-entropy
Batch size	256
Max epochs	20
Early stopping	Patience = 5 (validation loss)
Cross-validation	5-fold GroupKFold

Table 17: Pairwise AUC with bootstrap 95% CIs for four model-mode configurations (1,000 resamples). All boundary pairs (AUC $> \theta$) $p < 0.01$; OLMo 1v2–4v5 non-significant ($p > 0.09$).

	Qwen-rw	Pythia	OLMo	Qwen-cont
0v1	.996 [.995, .997]	.972	.986 [.984, .988]	.997 [.996, .998]
1v2	.696 [.686, .704]	.607 [.598, .617]	.500 [.496, .519]	.611 [.602, .621]
2v3	.589 [.580, .599]	.544 [.534, .554]	.497 [.486, .509]	.561 [.551, .571]
3v4	.544 [.533, .555]	.505	.503 [.491, .513]	.541 [.531, .551]
4v5	.534 [.524, .545]	.508	.497 [.483, .506]	.514 [.504, .526]

Table 18: Spearman ρ between binary detector scores and true depth, and 6-class accuracy. Correlations driven by 0v1; scores do not track depth beyond generation 1. Qwen-rw exhibits reversed sign ($\rho > 0$).

	Qwen-rw	Pythia	OLMo	Qwen-cont
Spearman ρ				
log-PPL (ref)	-0.123	-0.297	-0.378	-0.389
log-PPL (obs.)	$+0.082$	-0.453	-0.461	-0.162
PPL ratio	$+0.119$	-0.392	-0.407	-0.058
6-class acc.	0.199	0.130	0.127	0.183

Random baseline: 0.167

Table 19: Pythia-1.4B pairwise AUC under four decoding configs (10K bootstrap). †Greedy triggers repetition collapse (4-gram repeat: d_0=0.01 → d_1=0.72).

	Greedy†	p=0.9	T=0.7	T=1.2
0v1	.999	.995	.997	.993
1v2	.884	.677	.727	.617
2v3	.534	.555	.594‡	.523
3v4	.523	.530	.570	.517
4v5	.516	.501	.528	.505
$K^*_{\text{empirical}}$	2	2	2	2

‡CI upper 0.606 $> \theta$; point est. below.

Table 20: Pairwise AUC for three segment lengths (Qwen-base, cont. mode). $K^*_{\text{empirical}}$=2 at 128/256; 512 marginal (†CI lower $< \theta$).

	128	**256**	**512**
0v1	.979	.991	.992
1v2	.622	.628	.607†
2v3	.529	.556	.572
3v4	.535	.530	.530
4v5	.501	.514	.518
$K^*_{\text{empirical}}$	2	2	2†

Table 21: Pairwise AUC for three text domains (Qwen-cont). arXiv yields $K^*_{\text{empirical}}$=3; C4 and Wikipedia $K^*_{\text{empirical}}$=2.

	C4	**Wiki**	**arXiv**
0v1	.997	.996	.995
1v2	.611	.619	.858
2v3	.561	.550	.643*
3v4	.541	.523	.544
4v5	.514	.514	.529
$K^*_{\text{empirical}}$	2	2	3

*CI [.628, .658]; $p < 0.001$.

Table 22: Scorer sensitivity: $K^*_{\text{empirical}}$ under self- vs. cross-scoring. Cross-scoring reduces $K^*_{\text{empirical}}$ only for marginal Qwen-cont.

Generator	**Mode**	**Self**	**Pythia**	**OLMo**	**Agr.**
Qwen-Inst	Rewrite	2	2	2	✓
Qwen-Inst	Cont.	2	1	1	×
Qwen-base	Cont.	2	2	2	✓
Pythia	Cont.	2	2	2	✓
OLMo	Cont.	1	–	–	–
Qwen-7B	Cont.	2	–	–	–
Mistral	Cont.	1	–	–	–

B Extended Discussion

B.1 Theoretical Implications of $K^*_{\text{empirical}} = 2$

The DPI guarantees a finite K^*_{theory}; for general web text, $K^*_{\text{empirical}} \leq 2$ across all 10 configurations (1B–8B, 6 families), while structured domains (arXiv) yield $K^*_{\text{empirical}}$=3, aligning with model collapse convergence (Shumailov et al., 2024; Dohmatob et al., 2024; Alemohammad et al., 2024). Decoding strategy modestly affects signal strength (1v2 AUC $\in [0.6866, 0.7039]$) but does not shift the boundary.

B.2 Cross-Model Transfer Implications

Cross-model classification retains 80–100% accuracy with scorer-invariant feature rankings ($\rho > 0.90$), enabling single-classifier deployment across generators. Absolute discriminability at the $K^*_{\text{empirical}}$ boundary is scorer-sensitive: alternative scorers reduce Qwen-continuation $K^*_{\text{empirical}}$ from 2 to 1, while rewrite-mode $K^*_{\text{empirical}} = 2$ is preserved. Model family similarity modulates transfer (Pythia↔OLMo: 90–92% vs. Qwen-cont↔OLMo: 80–85%).

Table 23: Cross-scorer validation on Qwen-generated text. Alternative scorers (Pythia-1.4B, OLMo-1B) preserve feature ranking ($\rho > 0.90$) and rewrite-mode $K^*_{\text{empirical}} = 2$, but reduce continuation-mode $K^*_{\text{empirical}}$ from 2 to 1.

Scorer	Gen. Data	K^*	AUC(0v1)	AUC(1v2)	Spearman ρ
Qwen-1.5B (self)	Qwen-RW	2	—	0.6955	—
Pythia-1.4B	Qwen-RW	2	0.9923	0.7012	0.906
OLMo-1B	Qwen-RW	2	0.9918	0.6947	0.920
Qwen-1.5B (self)	Qwen-cont	2	—	0.6112	—
Pythia-1.4B	Qwen-cont	1†	0.9900	0.5923	0.942
OLMo-1B	Qwen-cont	1†	0.9890	0.5858	0.902
Qwen-1.5B	Qwen-7B-cont	2	0.9644	0.6178	—

Table 24: Three-way ANOVA (Generator × Scorer × Depth-Pair) on per-fold pairwise AUC values. DV = per-fold AUC; N=300 (12 cells × 5 folds × 5 depth-pairs). Scorer is statistically significant but explains modest variance (η_p^2=.166) relative to Depth-Pair (η_p^2=.997), supporting near-scorer-independence of depth discrimination.

Source	df	F	p	η_p^2
Generator	2	851.91	<.001	.866
Scorer	3	17.48	<.001	.166
Depth-Pair	4	22737.70	<.001	.997
Gen×Scorer	6	7.50	<.001	.146
Gen×Depth	8	913.41	<.001	.965
Scorer×Depth	12	2.25	.010	.093
Residual	264			

B.3 Feature Importance Across Decoding Strategies

Feature importance ordering varies across decoding strategies: greedy and T=0.7 share highly correlated rankings ($\rho = 0.914$), while greedy vs. T=1.2 yields lower correlation ($\rho = 0.53$, non-significant after Bonferroni). The depth boundary is robust but the optimal feature subset is strategy-dependent.

B.4 Scale and Neural Probe

Cross-scale validation confirms $K^*_{\text{empirical}} = 2$ from 1.5B through 8B: Qwen-7B (1v2 AUC = 0.6168 $[0.608, 0.626]$) and LLaMA-3.1-8B (1v2 AUC = 0.6123, 0v1 = 0.9224) both exceed θ, with progressively lower 0v1 signal consistent with larger models generating more human-like text. Gemma-2-27B (C4 continuation, Qwen cross-scored) yields $K^*_{\text{empirical}}$=1 (1v2 AUC = 0.5894), consistent with Gemma-2-2B at 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981). The Gemma family produces $K^*_{\text{empirical}}$=1 at both 2B and 27B, suggesting $K^*_{\text{empirical}}$ is model-family-dependent rather than scale-dependent.

The neural probe yields model-dependent $K^{*\text{neural}}_{\text{empirical}}$: 3 for Qwen (self-familiarity effect from instruction-tuned representations) but 2 for Pythia; OLMo yields $K^{*\text{neural}}_{\text{empirical}} = 1$ (1v2 AUC = 0.5054), consistent with the 15D LightGBM boundary. The 15D generator-agnostic features yield $K^*_{\text{empirical}} = 2$ for Pythia and Qwen, while OLMo yields $K^*_{\text{empirical}} = 1$ under the standardized pipeline.

B.5 LLM Zero-Shot Baseline

GPT-5.5 achieves 33.2% accuracy (MAE = 1.10) and Claude Opus 4.6 33.1% (MAE = 1.15) on a balanced 500-sample test set, comparable to OBD-LightGBM's 38.0% (95% CI: $[0.340, 0.422]$; McNemar $p = 0.081, 0.069$; Table 11). Per-depth recall confirms convergent failure at depths ≥ 2 (Section 6). When 6-class probabilities are marginalized to 5 classes ($P(d{\geq}4)$), accuracy rises to 44.4% but concentrates in depth-4 (recall: 22%→80%) while depths 2–3 collapse to 2–8%; we report raw 6-class results.

B.6 Practical Considerations

All filtering strategies degrade general capabilities (MMLU: 0.6018 → 0.5865–0.5941; HellaSwag: 0.6821 → 0.6638–0.6701); our 100% synthetic

setting is an upper bound; in mixed corpora, degradation scales with contamination fraction (Shumailov et al., 2024; Dohmatob et al., 2024). K^*-aware binary filtering reduces MMLU degradation by $\sim$50% (-1.53 pp $\rightarrow$ -0.77 pp), requires no per-model calibration, and is combinable with distribution-matching methods (DSIR, DoReMi).

B.7 Additional Scope Analysis

Prompt-chain versus retrain-chain. Retrain-chain validation (Pythia-1.4B LoRA, 5 generations) yields $K^*_{\text{empirical}} \geq 4$ (1v2 AUC $= 0.9484$, 2v3 $= 0.8888$, 3v4 $= 0.7982$), far exceeding prompt-chain $K^*_{\text{empirical}}=2$. The elevated signal arises from cumulative model-adaptation artifacts rather than text-intrinsic degradation; prompt-chain $K^*_{\text{empirical}}$ is the operationally relevant boundary for data curation.

Threshold sensitivity. At $\theta = 0.55$, $K^*_{\text{empirical}} = 3$ for Qwen-rewrite/continuation; at $\theta = 0.70$, $K^*_{\text{empirical}} = 1$ across all configurations (Figure 3). Qwen-continuation's 1v2 CI lower bound exceeds θ by only 0.2 pp; all ten boundary comparisons yield permutation $p < 0.001$.

Structural confound: token length. In continuation mode the generation prompt is discarded, so all depths ≥ 1 consist entirely of machine-generated tokens and the 1v2 boundary compares texts differing only in conditioning context (human-authored vs. machine-generated prompt), not structural composition. Depth 0 texts average $\sim$191 subword tokens (C4 truncation), while depth 1+ texts average 193–256 tokens depending on model (Table 13). This asymmetry is directionally opposite to a length-inflation confound: shorter depth-0 texts provide less feature-extraction context, making 0v1 discrimination harder, not easier. Within depths 1–5, mean token counts are stable (within-model range ≤ 4 tokens for all configurations except Qwen-continuation at 6 tokens), so the pairwise comparisons underlying $K^*_{\text{empirical}}=2$ are unaffected by length. EOS truncation rates are similarly depth-stable within each model: Qwen-rewrite 7–11%, Pythia 34–36%, OLMo $<$1%, Qwen-continuation 17–20%, Qwen-7B 45–47%. Models with the highest truncation (Pythia, Qwen-7B) still yield $K^*_{\text{empirical}}=2$ with the same step-function AUC decay pattern as low-truncation models (Table 2), confirming that EOS truncation does not drive the boundary.

Graduated filtering. All α values (0.1–0.9) underperform binary-ours on MMLU (α=0.1: 0.5873; α=0.9: 0.5892 vs. 0.5941); at α=0.9, HellaSwag matches exactly (0.6701). Binary-ours significantly outperforms all alternatives on MMLU ($p_{\text{corr}} \leq 0.021$); Graduated$_{0.9}$ is indistinguishable on HellaSwag ($p = 1.00$) but worse on MMLU ($p_{\text{corr}} = 0.011$).

B.8 Dose-Response: Contamination Ratio Effect

The main filtering experiments (Section 5) use 100% synthetic data to maximize effect size. To characterize behavior at lower contamination ratios, we mix synthetic depths 1–5 with clean C4 at 10%, 30%, and 50% contamination (Qwen-1.5B continuation, 5 seeds each; Table 25).

Table 25: MMLU accuracy by contamination ratio (fraction of synthetic data in training set) and filtering strategy. Base model (no fine-tuning): 0.6018. Mean $\pm$ std over 5 seeds.

Strategy	10%	30%	50%
No-filter	$0.6009 \pm .003$	$0.5978 \pm .004$	$0.5932 \pm .004$
Binary-ours	$0.6011 \pm .003$	$0.5988 \pm .003$	$0.5956 \pm .004$
Graduated	$0.6008 \pm .003$	$0.5982 \pm .004$	$0.5944 \pm .004$

At 10% contamination, no-filter MMLU (0.6009) remains within noise of the base model (0.6018, $\Delta = -0.09$ pp), and all three strategies are indistinguishable ($\Delta_{\max} = 0.03$ pp between strategies, $p > 0.8$). At 30%, differences emerge directionally ($\Delta = 0.10$ pp, binary vs. no-filter) but remain non-significant ($p > 0.3$). At 50%, binary-ours shows a 0.24 pp advantage over no-filter, approaching but not reaching significance ($p = 0.12$). The strategy ordering (binary $>$ graduated $>$ no-filter) is consistent with the 100% setting, but effect sizes scale linearly with contamination fraction. This demonstrates that K^*-aware curation provides practical benefit primarily when contamination is severe enough to cause measurable quality degradation.

B.9 Seed Robustness

Because the random seed governs both corpus sampling and generation quality (not merely CV splits), absolute AUC values are seed-dependent. We re-run three configurations with seed $= 123$ (primary: seed $= 42$; Table 26). $K^*_{\text{empirical}}$ is stable in 2/3 cases: Pythia-1.4B on Wikipedia yields $K^*_{\text{empirical}}=1$ under both seeds (1v2 $\Delta = -0.8$ pp), and Qwen-1.5B on C4 yields $K^*_{\text{empirical}}=2$ (1v2 $\Delta = +9.0$ pp, both seeds above θ). Qwen-1.5B on

Wikipedia shifts from $K^*_{\text{empirical}}$=2 to $K^*_{\text{empirical}}$=3 under seed = 123, but the 2v3 AUC (0.6067) is marginal: one of five folds falls below θ (fold 5 = 0.595), so this does not constitute strong evidence for $K^*_{\text{empirical}}$=3. The ±9 pp absolute AUC variation across seeds strengthens the methodological case for threshold-based $K^*_{\text{empirical}}$ over raw AUC comparisons: the qualitative boundary is robust even when point estimates shift.

Table 26: Seed robustness: $K^*_{\text{empirical}}$ under two random seeds. 1v2 Δ: change in 1v2 AUC from seed 42 to seed 123. $K^*_{\text{empirical}}$ is stable in 2/3 configurations; the Qwen-Wiki shift to $K^*_{\text{empirical}}$=3 is marginal (2v3 fold variance crosses θ).

Model	Domain	$K^*_{\text{empirical}}$		1v2 Δ	Stable?
		seed=42	seed=123		
Pythia-1.4B	Wikipedia	1	1	−0.8 pp	✓
Qwen-1.5B	C4	2	2	+9.0 pp	✓
Qwen-1.5B	Wikipedia	2	3†	+8.7 pp	×

†2v3 AUC = 0.6067; fold 5 = 0.595 < θ. Marginal boundary.

B.10 Bonferroni-Corrected Confidence Intervals

To control the family-wise error rate across all 10 model configurations, we apply Bonferroni correction: $\alpha_{\text{corr}} = 0.05/10 = 0.005$, yielding 99.5% confidence intervals ($z = 2.807$). Table 27 reports the corrected intervals for each configuration's $K^*_{\text{empirical}}$ boundary pair. Seven of ten configurations retain CI lower bounds above θ=0.60; three marginal $K^*_{\text{empirical}}$=2 configurations (Pythia-1.4B, Qwen-cont, LLaMA-8B) see their lower bounds dip below θ under the stricter criterion (0.5934, 0.5976, and 0.5988, respectively). These near-misses reflect narrow effect sizes (AUC − θ < 0.013) rather than classification failure: all three retain point estimates above θ, and Qwen-cont and LLaMA-8B retain uncorrected 95% CI lower bounds above θ (Pythia-1.4B falls marginally below at 0.5976).

B.11 Cross-Sample Robustness: 5K vs. 10K

The preceding analyses fix the sample pool at N=5,000 passages per depth. To assess between-sample variability, we re-draw an independent N=10,000 sample from C4 using a different random seed and repeat the full OBD pipeline. If $K^*_{\text{empirical}}$ and the boundary AUC are driven by the underlying distributional shift rather than by idiosyncrasies of a particular sample, both quantities should remain stable across draws.

Table 27: Bonferroni-corrected 99.5% confidence intervals for $K^*_{\text{empirical}}$ boundary pairs across all 10 configurations. $\alpha_{\text{corr}} = 0.005$; CI computed as AUC ± 2.807 × $\hat{\sigma}_{\text{boot}}$. "Pass θ": CI lower bound > 0.60.

Configuration	Boundary	AUC	$\hat{\sigma}_{\text{boot}}$	99.5% CI	Pass θ
$K^*_{empirical}$ = 2 configurations					
Qwen-rw (Instruct)	1v2	.6955	.0046	[.683, .708]	✓
Pythia-1.4B	1v2	.6074	.0050	[.593, .621]	×
Qwen-base	1v2	.6260	.0047	[.613, .639]	✓
Qwen-7B‡	1v2	.6168	.0046	[.604, .630]	✓
LLaMA-8B	1v2	.6123	.0048	[.599, .626]	×
Qwen-cont (Instruct)	1v2	.6112	.0049	[.598, .625]	×
$K^*_{empirical}$ = 1 configurations					
OLMo-1B	0v1	.9860	.0009	[.983, .989]	✓
Mistral-7B	0v1	.9209	.0010	[.919, .923]	✓
Pythia-6.9B§	0v1	.9599	.0015†	[.956, .964]	✓
Gemma-2-2B	0v1	.9530	.0018	[.948, .958]	✓

‡Generator: Qwen2.5-7B; scorer: Qwen2.5-1.5B.
§Nucleus sampling (p=0.95, T=1.0).
†Estimated from analogous configurations (no separate bootstrap JSON).

Table 28 reports the results. For Pythia-1.4B (continuation, C4), the 5K and 10K samples agree on $K^*_{\text{empirical}}$=2: the 0v1 AUC is nearly identical (0.9722 vs. 0.9712), and the boundary 1v2 AUC shifts from 0.6074 to 0.6175 (Δ = +0.010). This narrow shift is consistent with natural sampling variability across C4 subsets and does not alter the qualitative depth boundary. The 2v3 pair remains near chance in both draws (0.54, 0.54), confirming that depths ≥ 2 are indistinguishable regardless of sample. Qwen-1.5B (continuation, C4) replicates the same pattern ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6346). LLaMA-3.1-8B (continuation, C4) is the sole $K^*_{\text{empirical}}$=2 configuration where the 10K OOF AUC drops below θ (0.5880; pipeline AUC = 0.6148; Bonferroni 99.5% CI [0.5776, 0.5978]), yielding $K^*_{\text{empirical}}$=1 under the stricter criterion. OLMo-1B (continuation, C4) is the only configuration that yields $K^*_{\text{empirical}}$=1 at both 5K and 10K, with virtually identical depth profiles (0v1 AUC = 0.9871 vs. 0.9860; 1v2 AUC = 0.4984 vs. 0.5001); the 1v2 AUC at random level is consistent with Qwen cross-scoring, which places OLMo well below the discrimination boundary. Gemma-2-2B is the sole configuration where $K^*_{\text{empirical}}$ increases from 5K ($K^*_{\text{empirical}}$=1, 1v2 AUC = 0.5981) to 10K ($K^*_{\text{empirical}}$=2, 1v2 AUC = 0.6099), but the 10K boundary AUC exceeds θ by only 0.010, placing this model at the precise $K^*_{\text{empirical}}$=1/2 boundary.

B.12 Bootstrap $K^*_{\text{empirical}}$ Stability

To assess within-sample stability, we bootstrap-resample the evaluation data (1,000 iterations) and recompute $K^*_{\text{empirical}}$ for each resample across 10 configurations spanning four domains (Table 29).

Table 28: Cross-sample robustness: pairwise AUC and $K^*_{\text{empirical}}$ under 5K and 10K C4 samples drawn with different random seeds. All configurations use continuation mode. $K^*_{\text{empirical}}$ is determined by the θ=0.60 threshold on 1v2 AUC.

Model	N	AUC(0v1)	AUC(1v2)	AUC(2v3)	$K^*_{\text{empirical}}$
Pythia-1.4B cont	5K		(see Table 27)		2
Pythia-1.4B cont	10K	0.9712	0.6175	0.5364	2
Qwen-1.5B cont	5K		(see Table 27)		2
Qwen-1.5B cont	10K	0.9875	0.6346	0.5616	2
LLaMA-8B cont	5K		(see Table 27)		2
LLaMA-8B cont	10K	0.9282	0.5880†	0.5247	1†
OLMo-1B cont	5K		(see Table 27)		1
OLMo-1B cont	10K	0.9871	0.4984	0.5059	1
Gemma-2-2B cont	5K		(see Table 10)		1
Gemma-2-2B cont	10K	0.9534	0.6099†	0.5409	2†

†OOF AUC (pipeline AUC = 0.6148); Bonferroni 99.5% CI $[0.5776, 0.5978]$ does not pass θ.
‡Borderline: Bonferroni 99.5% CI $[0.600,\ 0.619]$ (percentile); CI lower 0.0002 below θ. Sole config where $K^*_{\text{empirical}}$ increases from 5K to 10K.

Eight of ten configurations yield identical $K^*_{\text{empirical}}$ in 100% of resamples (CI width 0). The two exceptions, Gemma-2-2B on C4 ($K^*_{\text{empirical}}$=1 in 92.7%, $K^*_{\text{empirical}}$=2 in 7.3%) and LLaMA-8B on Wikipedia ($K^*_{\text{empirical}}$=1 in 94.3%, $K^*_{\text{empirical}}$=2 in 5.7%), both sit at the $K^*_{\text{empirical}}$=1/2 boundary with 95% CIs of width 1, consistent with their marginal 1v2 AUC values. The $K^*_{\text{empirical}} \leq 2$ upper bound holds in all resamples across all configurations.

Bootstrap CIs capture within-sample stability (resampling from a fixed corpus), while the 5K-vs.-10K comparison (Table 28) captures between-sample variability (independent draws from the source distribution). Together they bound $K^*_{\text{empirical}}$ uncertainty from complementary angles: within-sample stability is high (8/10 at 100%), and between-sample shifts are limited to boundary configurations (LLaMA-8B, Gemma-2-2B).

Table 29: Bootstrap $K^*_{\text{empirical}}$ stability (1,000 resamples). Each row reports the fraction of resamples yielding each $K^*_{\text{empirical}}$ value and the resulting 95% CI. Eight of ten configurations are perfectly stable; two boundary cases show mild variability (CI width 1).

Config	Domain	$K^*_{\text{empirical}}$	$K^*_{\text{empirical}}$=1	$K^*_{\text{empirical}}$=2	$K^*_{\text{empirical}}$=3	95% CI
Qwen-1.5B	C4	2	0%	100%	0%	[2, 2]
OLMo-1B	C4	1	100%	0%	0%	[1, 1]
Pythia-1.4B	C4	2	0%	100%	0%	[2, 2]
LLaMA-8B	C4	2	0.1%	99.9%	0%	[2, 2]
Gemma-2B	C4	1	92.7%	7.3%	0%	[1, 2]
LLaMA-8B	Wiki	1	94.3%	5.7%	0%	[1, 2]
Mistral-7B	Wiki	1	0%	100%	0%	[1, 1]
LLaMA-8B	arXiv	3	0%	0%	100%	[3, 3]
OLMo-1B	arXiv	1	100%	0%	0%	[1, 1]
Qwen-1.5B	Code	2	0%	100%	0%	[2, 2]

B.13 Computational Cost

Table 30 reports per-document inference cost for the full OBD pipeline (Qwen2.5-1.5B-Instruct scorer, NVIDIA RTX PRO 6000, 1,000 documents, single-document processing without batching). The scorer log-probability pass dominates GPU time (20.6 ± 1.9 ms/doc); CPU feature extraction adds 23.4 ± 6.1 ms/doc; LightGBM inference is negligible (0.74 ± 0.10 ms/doc). Total latency is 44.7 ± 7.7 ms/doc (∼22 docs/sec), making the pipeline practical for corpus-scale filtering without specialized hardware. Batched scoring would reduce the per-document GPU cost further.

Table 30: Per-document inference latency for the OBD pipeline. Qwen2.5-1.5B-Instruct scorer, RTX PRO 6000, N=1,000 documents, no batching. Mean ± std.

Stage	Latency (ms/doc)
Scorer log-prob (GPU)	20.6 ± 1.9
Feature extraction (CPU)	23.4 ± 6.1
LightGBM inference	0.74 ± 0.10
Total	$\mathbf{44.7 \pm 7.7}$

B.14 DeBERTa Upper-Bound Comparison

To assess how much of the theoretical discrimination boundary the 15-feature OBD classifier captures, we fine-tune DeBERTa-v3-base on raw text as an upper-bound probe. Table 31 compares $K^*_{\text{empirical}}$ across three domains. DeBERTa matches OBD on C4 and Code ($K^*_{\text{empirical}}$=2) but extends $K^*_{\text{empirical}}$ to 4 on arXiv (3v4 AUC $= 0.624$, CI $[0.613, 0.635]$), one depth beyond the 15-feature boundary. The gap quantifies the cost of generator-agnostic features: one depth level on structured text, zero on general web text.

Table 31: $K^*_{\text{empirical}}$ comparison: 15-feature OBD vs. DeBERTa-v3-base (Qwen-1.5B generation, θ=0.60). DeBERTa extends $K^*_{\text{empirical}}$ by one depth on arXiv but not on C4 or Code.

	OBD (15D)		**DeBERTa**	
Domain	$K^*_{\text{empirical}}$	boundary AUC	$K^*_{\text{empirical}}$	boundary AUC
C4	2	0.626 (1v2)	2	0.705 (1v2)
arXiv	3	0.643 (2v3)	4	0.624 (3v4)
Code	2	0.738 (1v2)	2	0.808 (1v2)